Microeconomics I

Class 9 — Risk aversion measured; stochastic dominance; review

Pavel Kocourek

Universitat de Barcelona · Master in Economics

27 October 2026

Plan for today

Problem set 2 back; warm-up
Arrow–Pratt: how risk averse is she?
In pairs: r_A and r_R
Comparing risk aversion; insurance
First-order stochastic dominance (FOSD)
In pairs: two cumulative distribution functions (CDFs)
Second-order dominance; mean-preserving spreads
Break
Practice: Part I on one page; five problems in pairs
Summary; the Part I exam

Practicalities

- Problem set 2: returned in this class, graded
- Slides and reading: pkocourek.com/teaching/micro-1 · pkocourek@ub.edu

Warm-up: double every price (and, for x and v, wealth)

At $p = (2, 3)$, $w = 12$: $x = h(p, 6) = (3, 2)$, $v = 6$, $e(p, 6) = 12$.

Per pair, one minute: four values.

$x(4, 6, 24) = ?$

$v(4, 6, 24) = ?$

$h\Big((4, 6), 6\Big) = ?$

$e\Big((4, 6), 6\Big) = ?$

Class 4, slide 14; class 5 (v); class 6 (h, e). Mas-Colell, Whinston and Green (MWG) Def. 2.E.1, 3.D, 3.E.

Arrow–Pratt: $r_A(x) = -u''(x)/u'(x)$ and $r_R(x) = x\, r_A(x)$

$$r_A(x) = -\frac{u''(x)}{u'(x)}$$ absolute risk aversion (risks in euros)

$$r_R(x) = -\frac{x\, u''(x)}{u'(x)} = x\, r_A(x)$$ relative risk aversion (risks as shares of wealth)

u: the Bernoulli utility over wealth x (class 8).

$$u = \sqrt{x}\text{:}\quad r_A(x) = \frac{1}{2x},\quad r_R(x) = \tfrac{1}{2}$$

r_A falls with x. Class 8's two-coin game for Ana ($\sqrt{x}$; 64 euros if both coins land heads, else 0): risk premium 12 euros on the game alone, 3 euros with 36 sure euros more.

Why not u'' alone? $10\sqrt{x} + 3$ makes the same choices as $\sqrt{x}$ (class 8), yet has 10 times its u''. r_A and r_R are unchanged by any $\beta u + \gamma$, $\beta > 0$.

MWG Def. 6.C.3 and u'' alone, p. 190; Def. 6.C.4 (decreasing absolute risk aversion), p. 193; Def. 6.C.5, p. 194.

In pairs: r_A, r_R and the attitude of four Bernoulli functions

Per pair: three answers per row ($x > 0$). Three minutes.

	$u(x)$	$r_A(x)$	$r_R(x)$	attitude
(a)	$4\sqrt{x} + 1$			
(b)	$-e^{-x/50}$			
(c)	$-\dfrac{1}{x^2}$			
(d)	x^2			

Answers revealed in class. Try it before class.

MWG Example 6.C.4 (constant r_A), p. 191; Def. 6.C.5, p. 194.

More risk averse: r_A never lower, certainty equivalents never higher

u_2 is more risk averse than u_1 — three equivalent tests:

(i) $r_A(x, u_2) \geq r_A(x, u_1)$ for every x — at least as curved everywhere

(ii) $u_2 = \psi(u_1)$, ψ increasing and concave

(iii) $c(F, u_2) \leq c(F, u_1)$ for every F — in euros

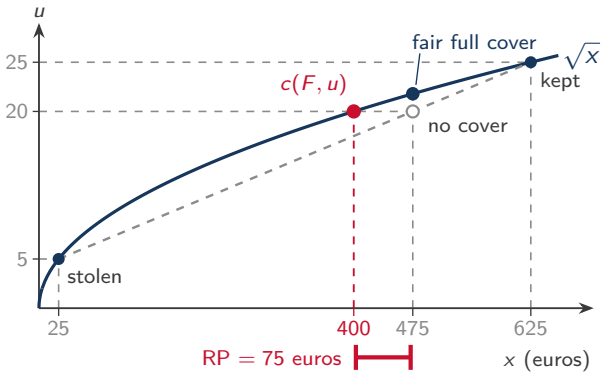
Certainty equivalent (class 8): the sure amount $c(F, u)$ with $u\big(c(F, u)\big) = \int u(x)\, dF(x)$.

A partial order: many pairs are not comparable.

Compare people by certainty equivalents, never by utility numbers.

MWG Prop. 6.C.2, p. 191: (i)–(v) are equivalent; (iv) and (v) are not needed here. Partial ordering, p. 192.

Insurance at a fair price: full cover



a 600-euro bike + 25 euros

stolen with probability $\pi = \frac{1}{4}$

no cover: 625 or 25 euros

$E = 475$, $c = 400$ euros;

risk premium $\mathrm{RP} = E - c = 75$ euros

fair cover, $q = \pi$ per euro:
600 euros of cover cost 150

475 euros for sure

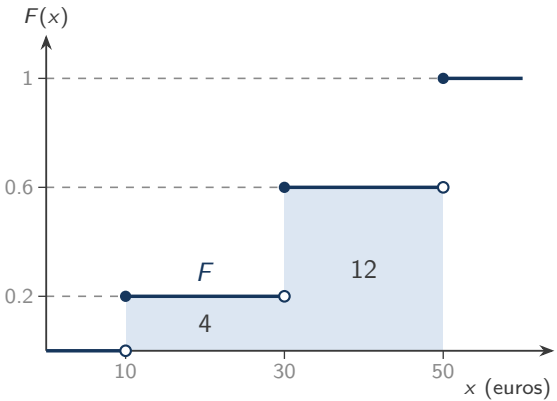
At a fair price every strictly risk-averse buyer insures fully (cover = the 600-euro loss). She would pay up to 225 euros for an expected payout of 150 euros.

MWG Example 6.C.1, pp. 187–188; price above π: partial cover, p. 189.

Comparing lotteries: stochastic dominance

MWG 6.D

A lottery's CDF: $F(x)$ = Prob(at most x); jumps are probabilities



F: 10, 30, 50 euros
with 0.2, 0.4, 0.4

$F(x) = \text{Prob}(\text{payoff} \leq x)$

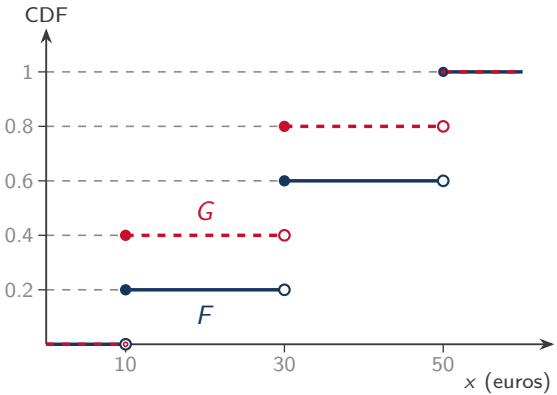
jump at 30 euros: $0.6 - 0.2 = 0.4$

area under F up to 50 euros:
$4 + 12 = 16$ euros
$= 50 - E[x] = 50 - 34$

Equal means $\Leftrightarrow$ equal areas under the two CDFs (from 0 to where both reach 1).

MWG 6.C, pp. 183–184, Fig. 6.C.1; (6.D.1), p. 198.

F dominates G (first order): $F(x) \leq G(x)$ at every x



first-order stochastic dominance (FOSD)
F FOSD G: $\int u\,dF \geq \int u\,dG$ for every nondecreasing u

everyone who likes money agrees

$\Leftrightarrow$ $F(x) \leq G(x)$ for every x

on	$[10, 30)$	$[30, 50)$	$[50, \infty)$
F	0.2	0.6	1
G	0.4	0.8	1

F never above G: everyone who likes money weakly prefers F (means 34, 26 euros).

MWG Def. 6.D.1 and Prop. 6.D.1, p. 195; Fig. 6.D.1, p. 196. G: 10, 30, 50 euros with 0.4, 0.4, 0.2.

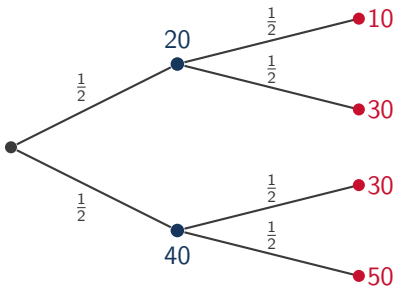
In pairs: two CDFs, P and Q

P: 10 or 50 euros, $\frac{1}{2}$ each Q: 20, 30, 60 euros with $\frac{1}{4}$, $\frac{1}{2}$, $\frac{1}{4}$

Per pair, one graph: compare interval by interval. Does either dominate? Three minutes.

Answers revealed in class.
Try it before class.

Same mean, more spread: every risk averter weakly prefers F



F G (euros)

second-order stochastic dominance (SOSD)
F SOSD G: same mean, and $\int u\,dF \geq \int u\,dG$ for every nondecreasing concave u

mean-preserving spread: each outcome of F becomes a lottery with that mean

G: 10, 30, 50 euros with $\frac{1}{4}$, $\frac{1}{2}$, $\frac{1}{4}$.
Both means: 30 euros.

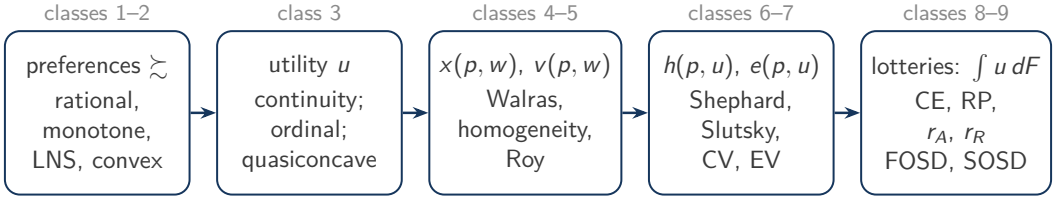
G is a mean-preserving spread of F, so F SOSD G.
Equivalent test, same mean: $\int_0^x G(t)\,dt \geq \int_0^x F(t)\,dt$ at every x (areas under the CDFs).

MWG Def. 6.D.2, p. 197; Example 6.D.2, pp. 197–198; (6.D.2), p. 198; Prop. 6.D.2, p. 199.

Break

Ten minutes

Part I on one page: from preferences to uncertainty



LNS: locally non-satiated · CV, EV: compensating and equivalent variation
CE: certainty equivalent

Part I = classes 1–9. Now five problems in pairs: try first, then the board.

MWG 1.B, 2.C–2.E, 3.B–3.G, 3.I, 6.B–6.D.

Demand, indirect utility, expenditure, Hicksian demand

A consumer has $u(x_1, x_2) = 2 \ln x_1 + x_2$, $x_1 > 0$.
Prices $p_1, p_2 > 0$, wealth $w > 2p_2$, so she buys both goods.
In pairs: one set of answers per pair.

(a) Walrasian demand $x(p, w)$. Answers revealed in class. Try it before class.

(b) Indirect utility $v(p, w)$.

(c) Expenditure function $e(p, u)$, by inverting v (for $u > 2 \ln(2p_2/p_1)$).

(d) $h_1(p, u)$ by Shephard's lemma; compare it with $x_1(p, w)$.

Classes 4–7: MWG 3.D, 3.E, 3.G (Shephard's lemma, Roy's identity).

Substitution and wealth effects at one point

Ines's Walrasian demand for good 1 is $x_1(p, w) = w\,p_2 \,/\, (p_1\,(p_1 + p_2))$ (two goods).
At $p = (1, 3)$ and $w = 32$; $h_1(p, u)$: her Hicksian demand for good 1, at $u = v(p, w)$.
In pairs: one set of answers per pair.

(a) Her demand x_1, and $\partial x_1/\partial p_1$. Answers revealed in class. Try it before class.

(b) The wealth effect,
$-x_1\,\partial x_1/\partial w$.

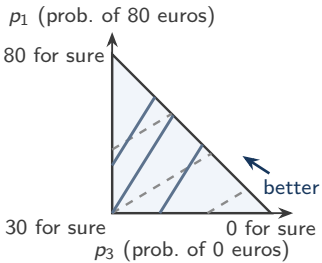
(c) The substitution effect
$\partial h_1/\partial p_1$ (Slutsky).

(d) At this point, which falls faster
as p_1 rises, x_1 or h_1? Why?

Classes 6–7: MWG 3.G (Hicksian demand, the Slutsky equation).

Indifference curves in the probability triangle

Prizes: 80, 30 and 0 euros. Petra maximizes expected utility. Solid: three of her indifference curves. Dashed: each line joins lotteries with the same expected value. Better lotteries lie up and to the left.



Which statement is true?

In pairs: a letter and a one-line reason.

A. She prefers 30 euros for sure to every other lottery with expected value 30 euros.

B. She prefers some lottery with expected value 30 euros to 30 euros for sure.

C. She is risk neutral: her curves are straight lines, like the dashed ones.

D. Nothing follows about her attitude to risk: her curves are not the dashed ones.

Answer revealed in class. Try it before class.

Class 8; MWG Fig. 6.B.5, p. 175; Def. 6.C.1, p. 185.

Dani chooses between two bonuses

Bonus F: 8 or 64 euros, $\frac{1}{2}$ each. Bonus G: 0, 16 or 64 euros, with $\frac{1}{4}$, $\frac{1}{4}$, $\frac{1}{2}$.
Dani's Bernoulli utility: $u(x) = x^{1/3}$.
In pairs: one set of answers per pair.

(a) Dani's $c(F, u)$, and $\mathrm{RP} = E[x] - c(F, u)$.

(b) Show that G is a mean-preserving spread of F.

(c) Use (b): which bonus does every risk averter prefer, and why?

Answers revealed in class. Try it before class.

MWG Def. 6.C.2, p. 186; Example 6.D.2 and Def. 6.D.2, pp. 197–198.

Who is more risk averse, Marta or Nil?

Marta and Nil maximize expected utility over wealth $x > 0$ (euros).
Marta: $u_M(x) = \sqrt{x}$. Nil: $u_N(x) = -1/x$. In pairs: one set of answers per pair.

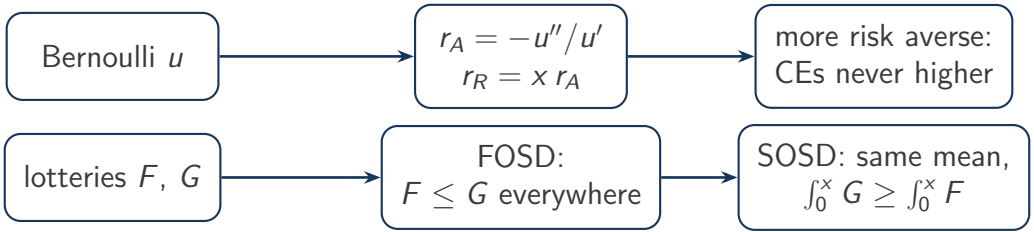
(a) $r_A(x)$ for Marta and for Nil. Answers revealed in class. Try it before class.

(b) Who is more risk averse? Check tests (i) and (ii) of slide 7.

(c) Wealth is 4 or 36 euros, probability $\frac{1}{2}$ each: Marta's and Nil's certainty equivalents.

MWG Def. 6.C.2, p. 186; Def. 6.C.3, p. 190; Prop. 6.C.2, p. 191.

Today in one line — and good luck on 28 October



Part I exam: Wednesday 28 October, Room 1030.

Arrive by 09:30; it starts at 09:35 sharp and ends at 11:20.

Bring photo ID and a pen. A simple calculator is allowed but never needed. Phones away.

Any handwritten notes, also as photocopies or printouts; no typed notes or other printed material.

Revise: Sextant, pkocourek.com/sextant/?c=micro-1

Part II (theory of the firm, general equilibrium) is taught by Olga Chiappinelli from November.

Thank you, and good luck.